

xMCF v3.1.1

ISO/PAS 8329

Description of mechanical connections
and joints in structural systems

ISO/PAS 8329 xMCF WG meeting
2025-11-26

Meeting results are
added in green font.

energizing great minds

Agenda

- 1 Short recap. Where are we today?
- 2 GitHub issues [#105](#), [#61](#), and [#25](#) (this one for cancellation?)
- 3 Additional discussion points — not applicable.
- 4 Arranging Ulrich Fox's succession — no time left for this agenda item.

Tomorrow is Thanksgiving in the US. That's why some team members can't participate today.
In particular, the discussion of the *questions wrt. Creo* is planned for the *next meeting*.



Short recap. Where are we today?

- In the last meeting, we set the *priorities for GitHub issues* based on the *voting results*.
- Accordingly, we will focus today on issues [#105](#) and [#61](#).
In view of their content, it makes sense to consider them together.



GitHub Issues — Top 6

<https://dud-poll.inf.tu-dresden.de/xMCF/> — as of 2025-11-12, 11:00 CET

Top 	Med 	#	Title	Volunteers
6	0	105	Support connection 0/1/2d to be reused (e.g. multiple occurrences)	CF, UF, NE, SH
4	2	61	xMCF shall Support Catalogs, e.g. of Standard Parts / Joining Elements	CF, UF, NE, JB
4	2	92	Setup a Document with Recommendations / Best Practices of xMCF Usage	CF, IR
4	2	95	Add persistent identifier to connection group	CF, UF, GH, NE, SH, TR
3	2	94	Add instance identifier to a connector	CF, UF, GH, NE, SH, TR
3	1	59	Mapping between xMCF and other established systems	CF, TR

Comments & Hints:

- JVH wrt. [#61](#): I think we also have a PDES user story from Penn for fastener catalog support in STEP?
- JVH wrt. [#59](#): There is also an *adhesives* symbology standard that is modeled similarly to the AWS.
- GH wrt. [#92](#): I attached public example CAD files to [#104](#), which could be useful for recommendations and their use cases.

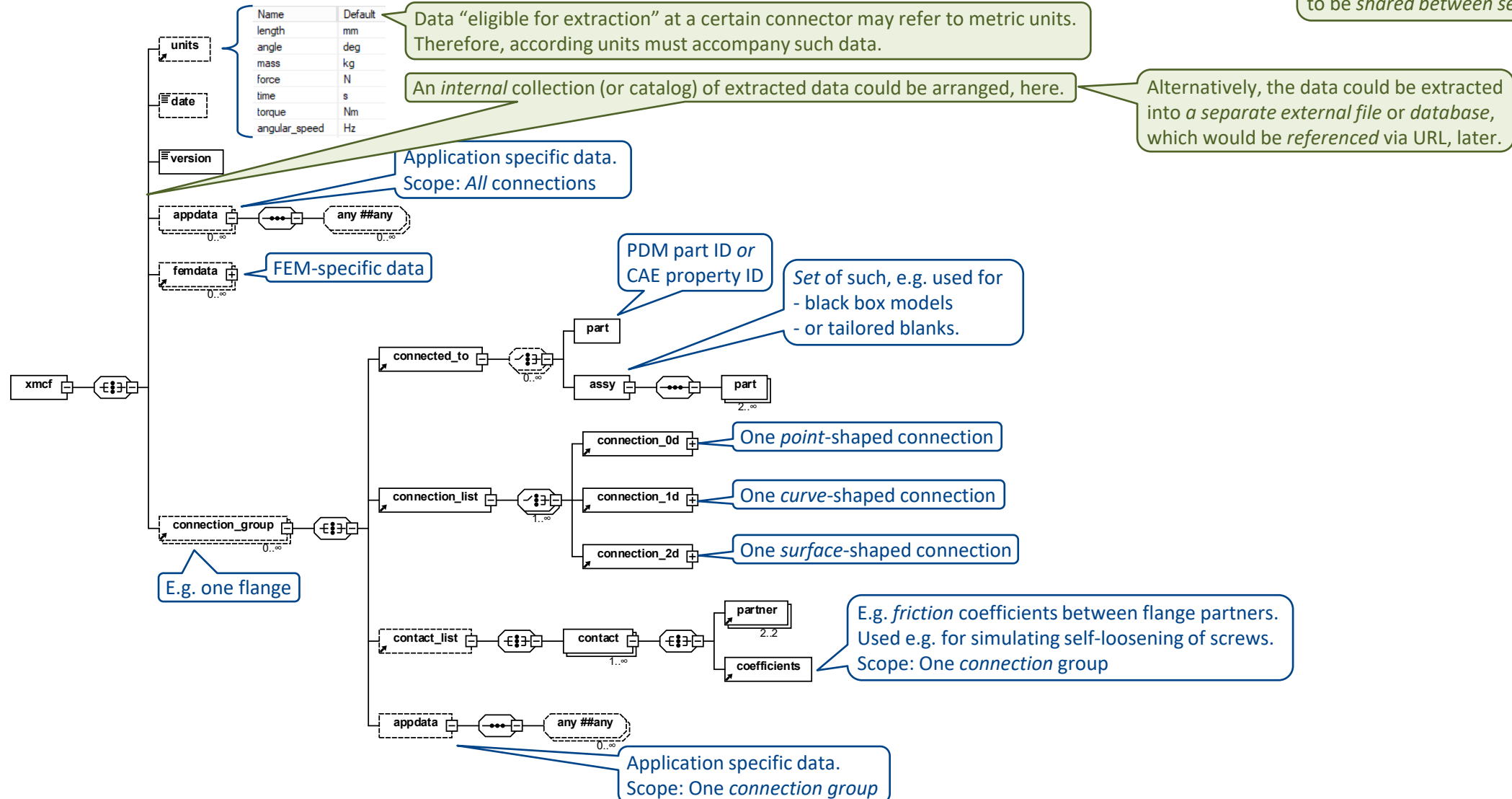


Issues #105 & #61

- Key statement of #105:
“The same rivet is used multiple times within one `connection_group`.
The only differences between these rivets are the 3D positioning (on each instance).
[...] enable to define the fastener only once and reuse it multiple time.”
 - Obviously, the same argument applies to other connection types, too. (It’s not “rivet specific”.)
- Key statement of #61:
“screw parameters such as length, diameter, head diameter, ... need to be described in detail for each screw.
Ditto for nuts, washers, rivets, [...]
Refer to a *catalog of standard parts* by a single, unique identifier, instead.”

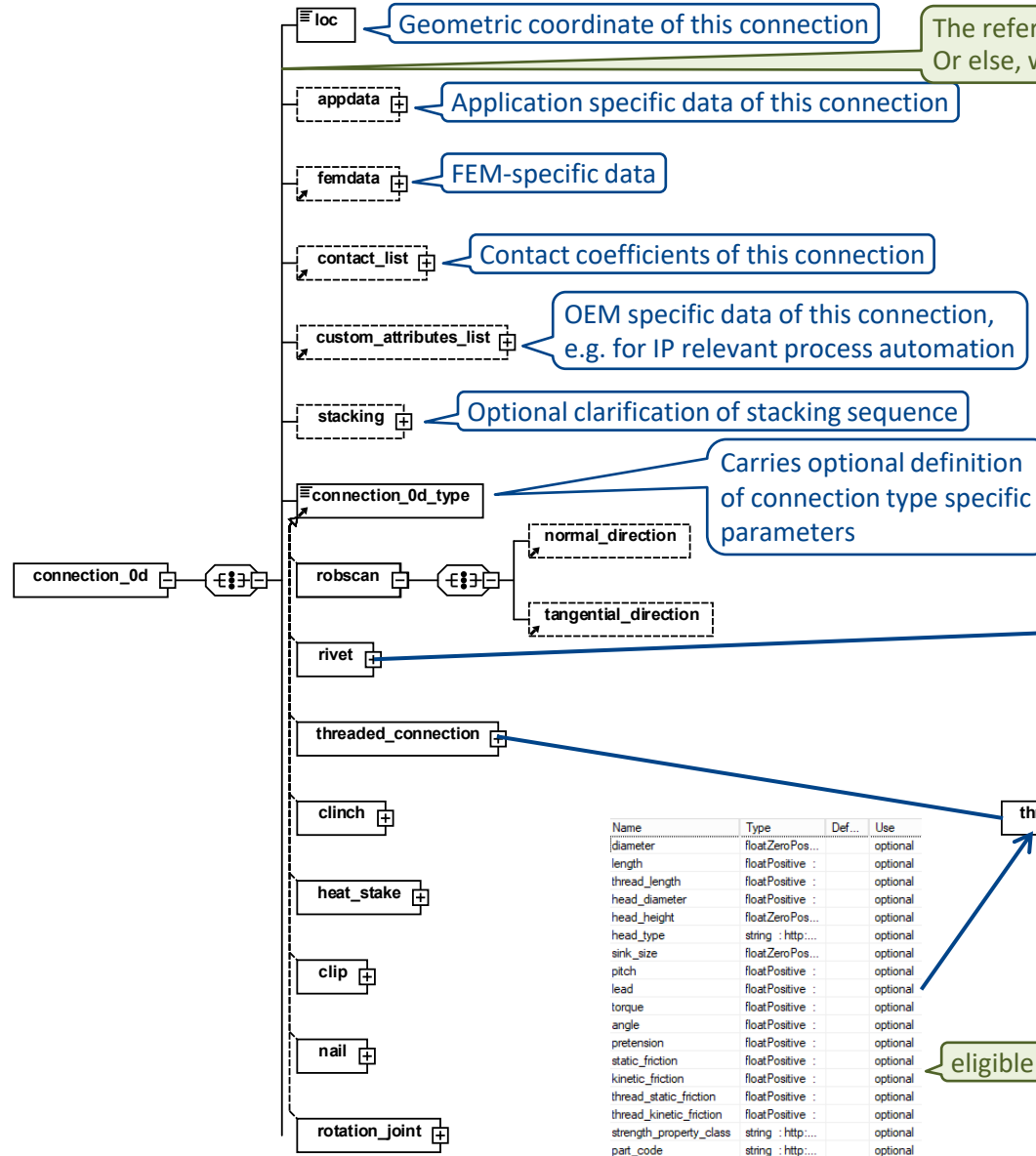
Let’s take a look at the xMCF data structure and see what we can do ...

Structure of χ MCF 3.1.1 XSD – High Level



Structure of xMCF 3.1.1 XSD – Point Connections

“eligible for extraction” to be understood as:
Data can be *taken out of one* specific connector,
to be *shared between several* connectors.



The reference to a catalog item could be arranged, here.
Or else, we use the `part_code`, where appropriate ...

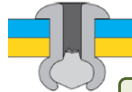
- For point connections, 8 types are supported. Each with its own, additional parameters.
- Some have sub-types and sub-elements, e.g. `<threaded_connection/>`.

eligible for extraction

Name	Type	Default	Use
hardness	floatPositive		optional
head_diameter	floatPositive		optional
head_height	floatZeroPositive		optional
head_type	string : http://ww...		optional
length	floatPositive		optional
part_code	string : http://ww...		optional
shaft_diameter	floatPositive		optional
sink_size	floatZeroPositive		optional
strength_property_class	string : http://ww...		optional

normal_direction

rivet_type



eligible for extraction

blind

self_piercing

solid

swop

clinch_rivet_stud

Name	Type	Default	Use
hole_depth	floatPositive		optional
hole_diameter	floatPositive		optional
max_grip	floatPositive		optional
min_grip	floatPositive		optional
shoulder_diameter	floatPositive		optional
shoulder_length	floatPositive		optional
tennon_diameter	floatPositive		optional
tennon_length	floatPositive		optional

normal_direction

threaded_connection_type

bolt

nut

washer

screw

flow_drilled

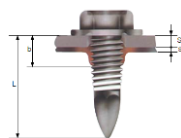
washer

Name	Type	Def...	Use
outer_diameter	floatPositive		optional
inner_diameter	floatPositive		optional
thickness	floatPositive		optional
attached	boolean : ht...	false	optional
static_friction	floatPositive		optional
kinetic_friction	floatPositive		optional
strength_property_class	string : http...		optional
part_code	string : http...		optional

Name	Type	t	Use
pre_machined_hole_diameter	floatZeroPos...		optional
pre_machined_hole_index	partIndex		optional
pilot_hole_diameter	floatZeroPos...		optional

eligible for extraction

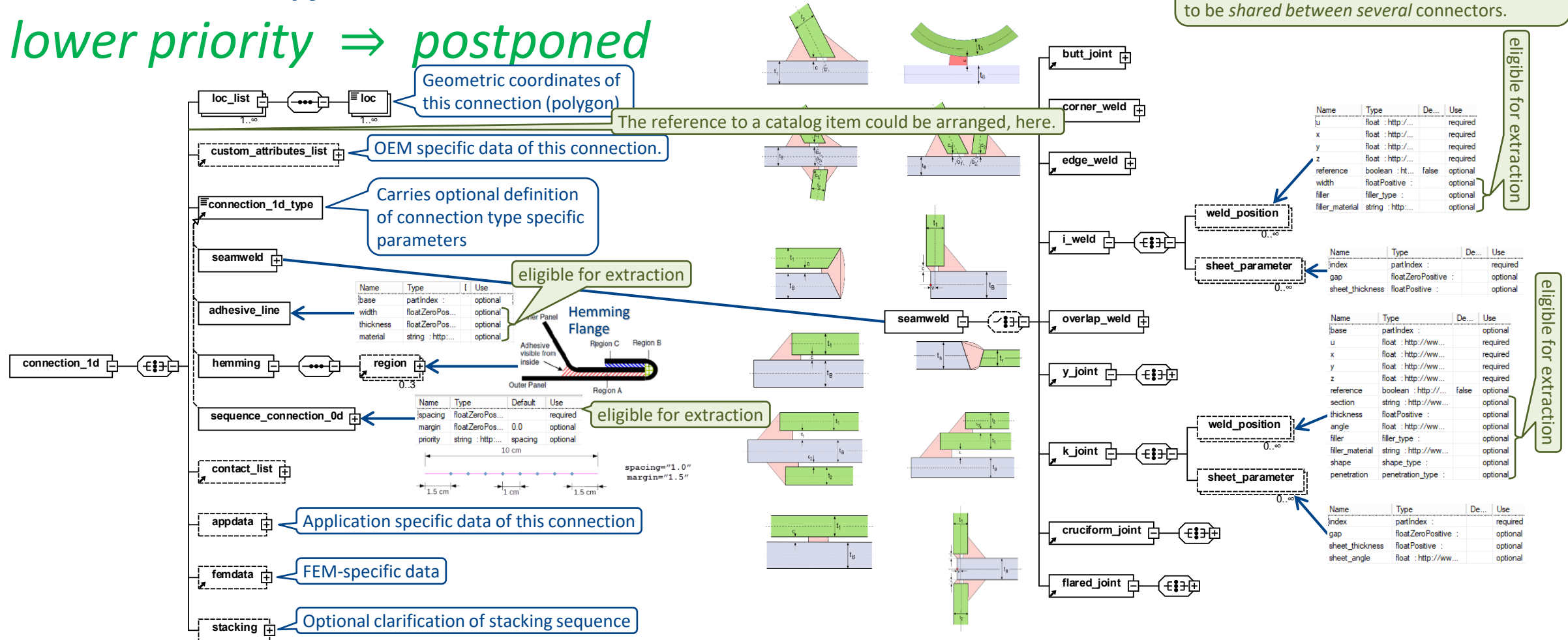
eligible for extraction



Structure of χ MCF 3.1.1 XSD – Curve Connections

lower priority $\Rightarrow$ postponed

“eligible for extraction” to be understood as:
Data can be *taken out of one* specific connector,
to be *shared between several* connectors.

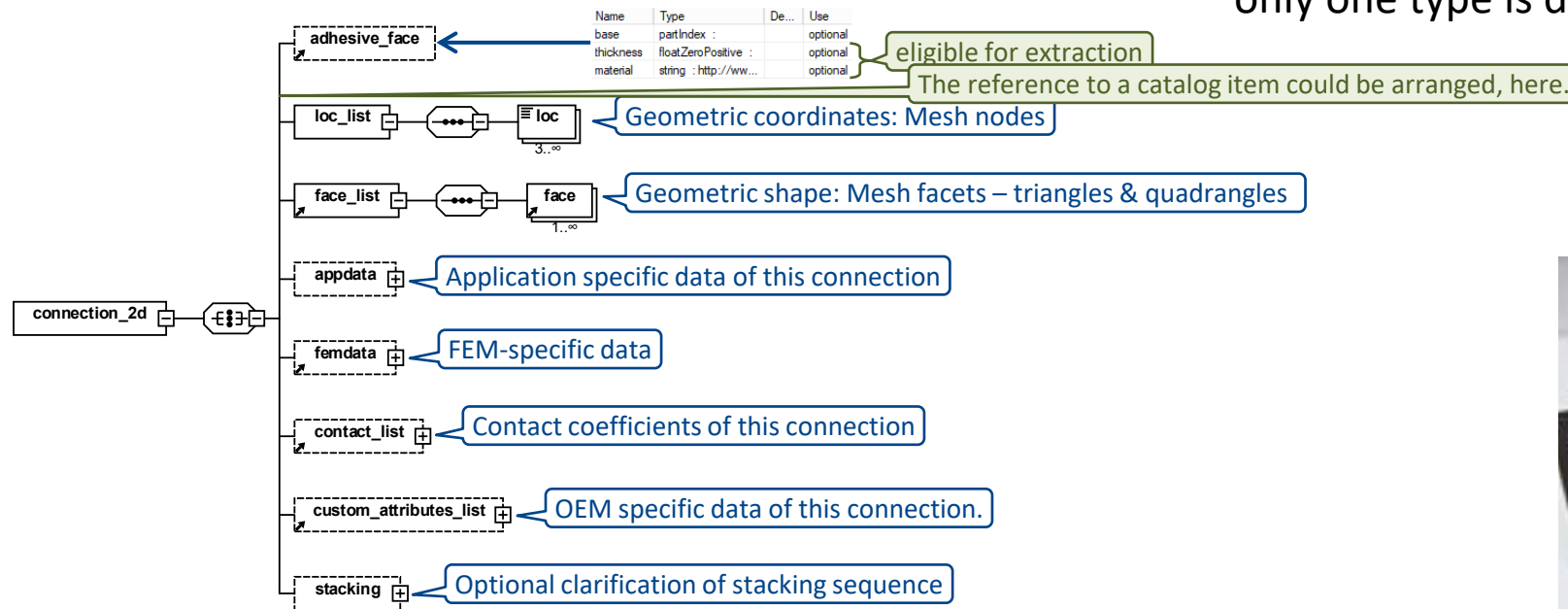


Structure of χ MCF 3.1.1 XSD – Surface Connections

lower priority $\Rightarrow$ postponed

“eligible for extraction” to be understood as:
Data can be *taken out of one* specific connector,
to be *shared between several* connectors.

- For surface connections, only one type is defined: adhesive gluing.



Questions & Considerations ... Starting points for more discussions

1. What happens, if a connector contains both, a reference to extracted data (catalog) *and* (some of) such data, internally?
Which data takes precedence?
“Internal” data shall take precedence.
 2. May a connector omit *mandatory* attributes if they can be determined via the reference to a catalog entry?
Yes, but schema validation must take this into account.
 3. Should a catalog entry be allowed to refer to other entries, maybe even in another catalog?
Could be used to enable a *step-by-step completion*.
Not allowed, for internal catalogs, for the moment.
Format of database-catalogs is not subject of the standard, therefore no statement given here.
 4. Should we allow both, *internal* & *external* catalogs?
Even in combination?
Yes.
- Tools need to store, where data “eligible for extraction” originally was read from (individual connector, internal or external catalog).
 - For new connectors, users must be able to specify and for existing ones, users must be able to modify this. In both cases, incl. the URL of the external catalog.
 - Should the system allow the modification of data originating from an external catalog?
Yes, but this creates *internal* data.
 - When writing an xMCF file:
Should the system emit a warning if data in an *external* catalog has changed, meanwhile?
Or, if the external catalog can no longer be found. (To avoid lost or inconsistent data, in the end.)
References to external databases must consider life cycle (e.g. release codes) of the items contained therein.
However, we can leave this responsibility to the process owner. (Can be discussed further, later.)

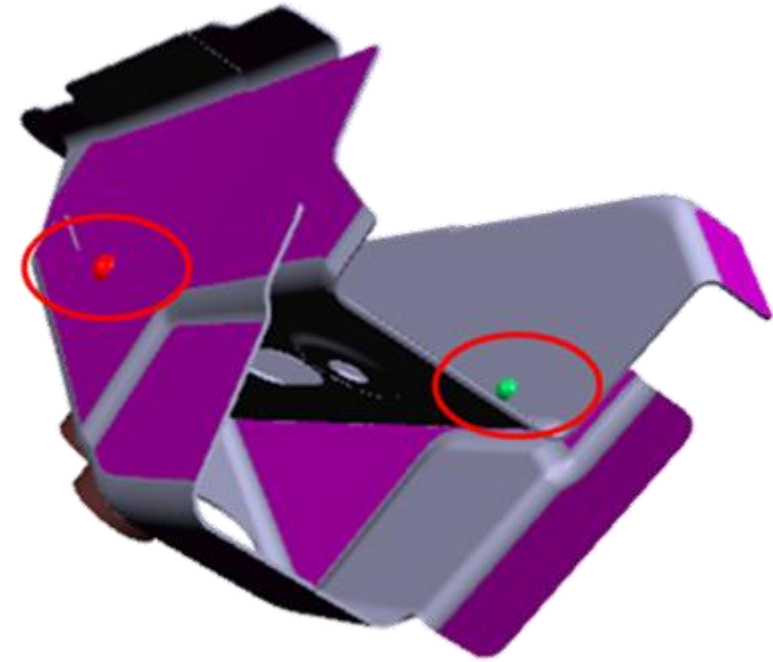
Other ISO Standards

[#61](#) suggests to analyse existing standards in this context, such as

- [ISO 13584-511:2006 — Industrial automation systems and integration — Parts library — Part 511: Mechanical systems and components for general use — Reference dictionary for fasteners](#)
- [ISO 1891:2009 — Fasteners — Terminology](#)
- Any more?
- Organizational question: How can we get copies of such standards?
⇒ Homework for all: Please see if you can find some useful information “at home”!

Guillaume's questions

- Catalogue, as identified using [VDA231-200](#) used at BMW: standard type (DIN EN, GS, SAE, ...), standard number, legal owner (especially important for internal company standards), issue date → **May be mapped to components of the URL?**
- `part_code`: At BMW, the welding points (spotwelds) are represented by a small sphere, which has a part number (see graphic) (not for logistic purposes, since it cannot be ordered or manufactured) → **We should transport this code. Attribute name to be discussed, though.**
- Potential attributes dedicated to each *individual* usage of a fastener: location/orientation, *instance number* (see [#94](#)), *tightening torque*. → **To be considered!**
- Difference between `connection_[012]d.label` and `part_code`? → See issue [#104](#). → **Maybe, formulation in standard document needs improvement. Please leave a hint in [#104](#), in this case.**



Ulrich Fox's succession as co-lead of ISO 8329/χMCF working group

Job description:

- Facilitate monthly core team meetings
- Facilitate quarterly full forum meetings
- Contribute and co-lead project management
- Contribute to discussion and resolution of issues and requirements
- Contribute to technical documentation

Average time consumption 2h/week

- To be resumed at the next meeting.

Next meetings are (to be) scheduled ...

- In 2 weeks, Dec. 10, 15:00 CET
Focus topics:
 - Recent progress & next steps with respect to
 - [#105](#) — [Support connection 0/1/2d to be reused \(e.g. multiple occurrences\)](#)
 - [#61](#) — [xMCF shall Support Catalogs, e.g. of Standard Parts / Joining Elements](#)
 - xMCF-related findings in PTC Creo
 - Planning Kickoff 2026
 - Arranging Ulrich Fox' succession
- ...

